

Covariance

This calculates the covariance of two sets of numeric data. The covariance is a measure of how much two random variables change together. To perform covariance, choose *Data > Statistics > Covariance*.

Data

Input range: Refers to the range of the data to analyse.

Results to: Refers to the top left cell of the range where the results will be displayed.

Grouped by: Selects whether the input data has a *columns* or *rows* layout.

An example and results are shown in Figure 5.

Exponential smoothing

Exponential smoothing is a filtering technique, which when applied to a data set, produces smooth results. It is employed in many domains such as the stock market, economics and in sampled measurements.

To perform exponential smoothing, choose *Data > Statistics > Exponential smoothing*.

Data

Input range: This refers to the range of the data to analyse.

Results to: This refers to the top left cell of the range where the results will be displayed.

Grouped by: Selects whether the input data has a *columns* or *rows* layout.

Parameters

Smoothing factor: A parameter between 0 and 1 represents the damping factor Alpha in the smoothing equation.

An example and results are given in Figure 6.

Moving average

This calculates the moving average of a time series. For the moving average, choose *Data > Statistics > Moving average*.

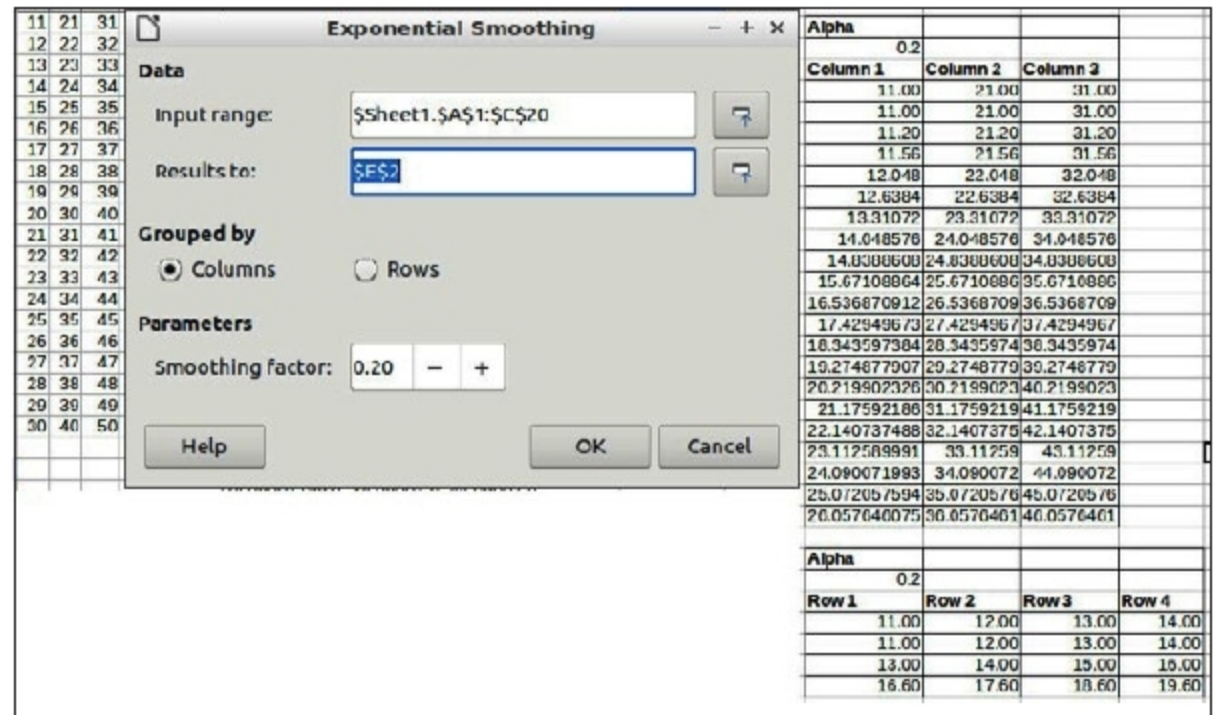


Figure 6: Sample data set, exponential smoothing options and results

Data

Input range: This refers to the range of the data to analyse.

Results to: This refers to the top left cell of the range where the results will be displayed.

Grouped by: Selects whether the input data has a *columns* or *rows* layout.

Parameters

Interval: This is the number of samples used in the moving average calculation. An example and results are shown in Figure 7.

Regression

In statistical modelling, regression

analysis is a set of statistical processes for estimating the relationships between a dependent variable (often called the 'outcome variable') and one or more independent variables (often called 'predictors', 'covariates' or 'features'). The most common form of regression analysis is linear regression, in which the researcher finds the line (or a more complex linear function) that most closely fits the data according to a specific mathematical criterion. The other two types of regression that can be performed are logarithmic and power regression.

Here, two columns are required. Sample data is used here for imaginary time and temperature data.

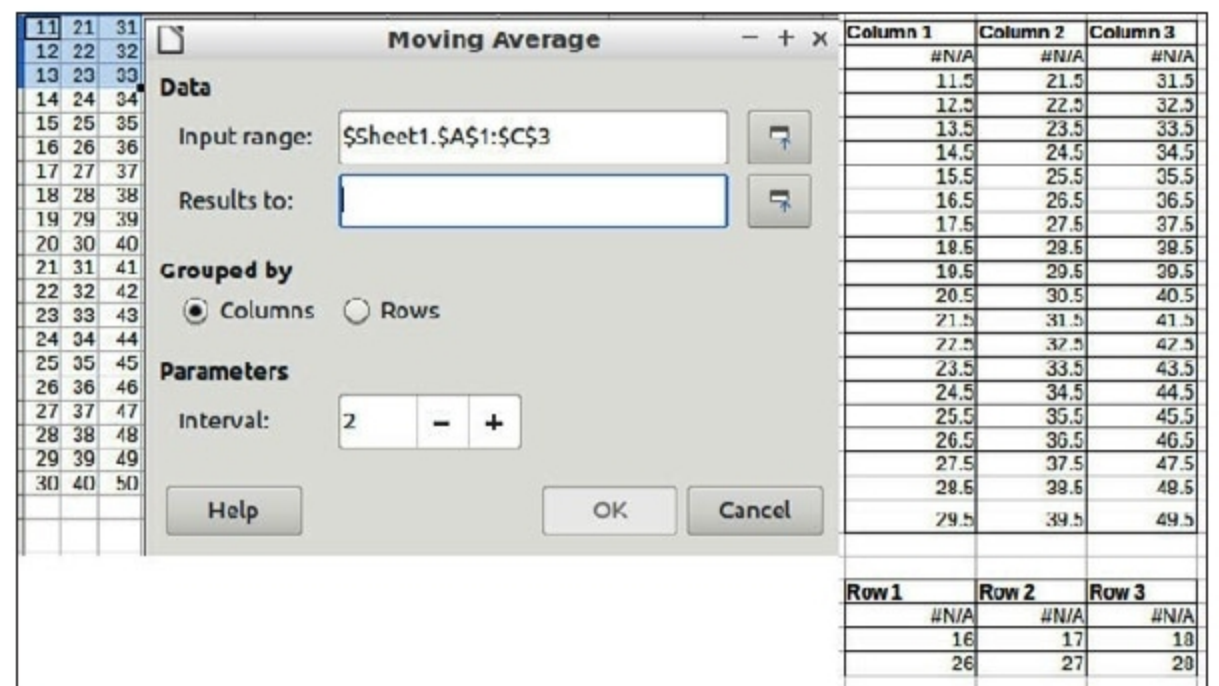


Figure 7: Sample data, moving average options and results